PK-PD Practice 2

Module 7b - PK-PD Relationship · Daily practice sheet · No date

Name _____ Date _____

What is the expected teff in a 165-pound male for a single IV dose of an antibiotic that is dosed at 5 mg/kg and has a minimum effective concentration of 12 mcg/mL?

(1st question, unnumbered)

The drug follows a one-compartment open model and has an apparent volume of distribution that is 0.15L/kg and elimination half-life of 4.75 hours.

(2nd question, unnumbered)

The drug follows a one-compartment open model and has an apparent volume of distribution that is 0.15L/kg and elimination half-life of 4.75 hours. What dose would you recommend to attain a teff of 12 hours? (Give the dose per kilogram.)

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

(1st question, unnumbered) Her key

Answer: 7 hr

WHAT THE STEM GIVES

weight	165-pound	÷ 2.2 = 75 kg
D_0	5 mg/kg	× 75 kg = 375 mg
Ceff	12 mcg/mL	equals 12 mg/L, the MEC
V_D	0.15L/kg	× 75 kg = 11.25 L
t½	4.75 hours	k = 0.1459 hr^{-1}

THE WORKING, ONE LINE AT A TIME

UNIT

$165\ lb \div 2.2 = 75\ kg$; $D_0 = 5\ mg/kg \times 75\ kg = 375\ mg$; $V_D = 0.15\ L/kg \times 75\ kg = 11.25\ L$

Both the dose and the volume are per kilogram, so the weight in pounds is turned into kilograms first and each is multiplied by 75 kg.

SETUP

$$k = \frac{0.693}{4.75\ hr} = 0.1459\ hr^{-1};\ Ceff = 12\ mcg/mL = 12\ mg/L$$

Her key uses the k form of the duration line, so the 4.75-hour half-life is turned into k first. 1 mcg/mL equals 1 mg/L, so the MEC is 12 mg/L.

ALGEBRA

$$teff = \frac{\ln(375 \div 11.25 \div 12)}{0.1459\ hr^{-1}} = \frac{\ln(33.33 \div 12)}{0.1459} = \frac{\ln 2.778}{0.1459} = \frac{1.022}{0.1459\ hr^{-1}}$$

C_0 = 375 mg ÷ 11.25 L = 33.33 mg/L, the level just after the bolus, and it has to fall 2.78-fold to reach the MEC of 12 mg/L before the effect ends.

ROUND

$teff = 7.0\ hr$

1.022 ÷ 0.1459 = 7.00 hr; her key boxes 7 hr. It is between one and two half-lives of 4.75 hr, because 2.78 lies between 2 and 4.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0 \div Ceff = 2.778$ is between 2 and 4, so teff is between one and two half-lives of 4.75 hr: 7.0 hr.

(2nd question, unnumbered) Her key

Answer: 10.4 mg/kg

WHAT THE STEM GIVES

teff	12 hours	the target duration, in the exponent
weight	165-pound	÷ 2.2 = 75 kg

Ceff	12 mcg/mL	equals 12 mg/L
V_D	0.15L/kg	× 75 kg = 11.25 L
t½	4.75 hours	k = 0.1459 hr^{-1}

THE WORKING, ONE LINE AT A TIME

UNIT

$$165 \text{ lb} \div 2.2 = 75 \text{ kg}; \; V_D = 0.15 \text{ L/kg} \times 75 \text{ kg} = 11.25 \text{ L}; \; k = \frac{0.693}{4.75 \text{ hr}} = 0.1459 \text{ hr}^{-1}$$

The dose line needs the patient's volume in L and k in hr^{-1}: the weight scales the per-kilogram volume, and the half-life gives k.

ALGEBRA

$$C_0 = Ceff \, e^{k \, teff} = (12 \text{ mg/L}) \, e^{(0.1459)(12)} = 12 \text{ mg/L} \times 5.759 = 69.11 \text{ mg/L}$$

For the level to be at the MEC of 12 mg/L at 12 hours, the level at time zero must be $e^{k \, teff}$ times higher. 12 mcg/mL is 12 mg/L.

ALGEBRA

$$D_0 = C_0 \times V_D = 69.11 \text{ mg/L} \times 11.25 \text{ L} = 777.5 \text{ mg}$$

The bolus that gives that C_0 is C_0 times V_D, because $C_0 = D_0$ over V_D for an IV bolus; mg/L × L leaves mg, the dose for this 75-kg patient.

ROUND

$$\frac{777.5 \text{ mg}}{75 \text{ kg}} = 10.4 \text{ mg/kg}$$

Her key gives 777.5 mg and boxes 10.4 mg/kg, the same per-kilogram form the stem doses in: about twice the 5 mg/kg that lasts 7 hr.

> **DOES THE NUMBER MAKE SENSE?**
> 12 hr is 5 hr longer than the 7 hr of 5 mg/kg, about one more half-life of 4.75 hr, so the dose roughly doubles: 10.4 mg/kg.